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CORE MODULES AND EARPHONES

Abstract

The present disclosure provides a core module and an earphone. The core module comprises a housing, a battery assembly, and a driving assembly. The housing is provided with a first accommodation space and a second accommodation space. The battery assembly is accommodated in the first accommodation space along a first preset assembly direction. The driving assembly is accommodated in the second accommodation space along a second preset assembly direction. The first preset assembly direction and the second preset assembly direction are provided at an acute angle. A first limiting portion is disposed in the first accommodation space. A projection of the first limiting portion on a cross section perpendicular to the first preset assembly direction does not overlap with a projection of the battery assembly on the cross section perpendicular to the first preset assembly direction. The first limiting portion is configured to limit a position of the battery assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Patent Application No. PCT/CN2023/131093, filed on Nov. 10, 2023, which claims priority to Chinese Patent Application No. 202310956970.8, filed on Jul. 28, 2023, the entire contents of each of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of electronic equipment, and in particular to core modules and earphones.

BACKGROUND

[0003] With the increasing popularity of electronic devices, electronic devices have become indispensable tools for social interaction and entertainment in people's daily lives, leading to increasingly higher demands for their performance. Devices such as earphones and smart glasses are now extensively used in everyday scenarios, often paired with mobile phones, computers, and other terminal devices to deliver an immersive auditory experience. Therefore, improving the sound quality of earphones and similar electronic devices has become a pressing issue that needs to be addressed.

SUMMARY

[0004] One embodiment of the present disclosure provides a core module, comprising: a housing, a battery assembly, and a driving assembly. The housing may be provided with a first accommodation space and a second accommodation space. The battery assembly may be accommodated in the first accommodation space along a first preset assembly direction. The driving assembly may be accommodated in the second accommodation space along a second preset assembly direction. The first preset assembly direction and the second preset assembly direction may be provided at an acute angle. A first limiting portion may be disposed in the first accommodation space. A projection of the first limiting portion on a cross section perpendicular to the first preset assembly direction may not overlap with a projection of the battery assembly on the cross section perpendicular to the first preset assembly direction. The first limiting portion may be configured to limit a position of the battery assembly.

[0005] In some embodiments, the first limiting portion is configured to stop the battery assembly at least in a direction perpendicular to the first preset assembly direction to limit rotation of the battery assembly around the first preset assembly direction.

[0006] In some embodiments, the battery assembly has a thickness direction, and an angle between the first preset assembly direction and the thickness direction of the battery assembly is in a range of 0°-5°.

[0007] In some embodiments, the battery assembly includes a top wall surface and a bottom wall surface which are spaced apart along the thickness direction and a peripheral wall surface connecting the top wall surface and the bottom wall surface. The first limiting portion may be configured to abut against the peripheral wall surface of the battery assembly for stopping.

[0008] In some embodiments, the first accommodation space includes a second limiting portion. The second limiting portion may be configured to limit movement of the battery assembly along a radial direction of the battery assembly.

[0009] In some embodiments, the first accommodation space includes a first inner sidewall. The second limiting portion may be a portion of the first inner sidewall, and/or the second limiting portion may be a limiting protrusion disposed on the first inner sidewall.

[0010] In some embodiments, a positioning portion is disposed in the first accommodation space. The positioning portion and the first limiting portion may be spaced apart. The battery assembly may be provided with a positioning protrusion, and the positioning protrusion may be disposed between the positioning portion and the first limiting portion along the first preset assembly direction, so that the positioning portion and the first limiting portion perform assembly positioning on the battery assembly when the battery assembly is assembled in the first accommodation space along the first preset assembly direction.

[0011] In some embodiments, the core module includes an electrode member electrically connected with the battery assembly. The electrode member may be disposed in the positioning portion and extend from the first accommodation space to outside of the housing.

[0012] In some embodiments, two positioning portions are provided, and the two positioning portions are respectively disposed on two sides of the first limiting portion, and the positioning protrusion is disposed between the first limiting portion and any one of the two positioning portions.

[0013] In some embodiments, the battery assembly leads a wire, the first accommodation space includes a first bottom wall, and a lead wire groove is disposed on the first bottom wall to accommodate the wire.

[0014] In some embodiments, the housing is provided with a partition wall. The partition wall may be configured to separate the first accommodation space and the second accommodation space. The first limiting portion may be convexly disposed on a side of the partition wall facing the first accommodation space.

[0015] In some embodiments, the housing includes a first shell and a second shell. The second shell may be provided with an open end. The first shell and the second shell may be connected with each other and the first shell may cover the open end to form the first accommodation space and the second accommodation space. The second shell may be provided with a plurality of positioning grooves. The first shell may be provided with a plurality of positioning columns corresponding to the plurality of positioning grooves. At least a portion of the plurality of positioning columns may be embedded in the plurality of positioning grooves. Notches of the plurality of positioning grooves may be provided with first inclined wall surfaces, and the first inclined wall surfaces may be configured to guide the plurality of positioning columns to be embedded into the plurality of positioning grooves.

[0016] In some embodiments, the plurality of positioning columns are configured to be formed along a first demolding direction of the first shell. The core module may comprise a circuit board, and the first demolding direction may be perpendicular to a plane where the circuit board is located.

[0017] In some embodiments, an angle between the first demolding direction and the first preset assembly direction is in a range of 0° - 5°

[0018] In some embodiments, the first inclined wall surfaces are inclined compared to a second demolding direction and are biased toward the first demolding direction.

[0019] In some embodiments, the first shell is provided with a first support platform, and the plurality of positioning columns are disposed on the first support platform and extend outward from a bottom surface of the first support platform. The second shell may be provided with a second support platform, and the plurality of positioning groove may be disposed on the second support platform and form the notches on a top surface of the second support platform. When the at

least a portion of the plurality of positioning columns are embedded in the plurality of positioning grooves, a first gap may be formed between the bottom surface of the first support platform and the top surface of the second support platform. A second gap may be provided between edges of the first shell and the second shell which are buckled with each other. A minimum gap height of the first gap may be less than a minimum gap height of the second gap, and the second gap may be configured to accommodate a fixing adhesive.

[0020] In some embodiments, a minimum gap distance of the first gap is in a range of 0.02-0.07 mm, and a minimum gap distance of the second gap is in a range of 0.08-0.12 mm.

[0021] One embodiment of the present disclosure provides an earphone comprising any core module described above and an ear hook configured to be connected with the housing to make the core module be worn near an ear canal of a user but not block the ear canal.

[0022] The beneficial effects of the present disclosure include the following content. The present disclosure provides the core module comprising the housing, the battery assembly, and the driving assembly. The housing is provided with the first accommodation space and the second accommodation space. The battery assembly is accommodated in the first accommodation space along the first preset assembly direction. The driving assembly is accommodated in the second accommodation space along the second preset assembly direction. The first preset assembly direction and the second preset assembly direction are provided at the acute angle. The first limiting portion is disposed in the first accommodation space. The projection of the first limiting portion on the cross section perpendicular to the first preset assembly direction does not overlap with the projection of the battery assembly on the cross section perpendicular to the first preset assembly direction. The first limiting portion is configured to limit the position of the battery assembly. Based on the above, the first limiting portion is arranged on one side of the battery assembly along the direction perpendicular to the first preset assembly direction. In this way, the first limiting portion abuts against the peripheral side wall of the battery assembly parallel to the first preset assembly direction (the specific descriptions of the peripheral wall surface can be found in the present disclosure below), so that the first limiting portion can effectively limit the movement of the battery assembly relative to the housing in a plane perpendicular to the first preset assembly direction, which effectively improves the stability of the connection between the battery assembly and the housing assembly, and effectively prevents aggravating the vibration of the earphone during operation caused by the relative movement of the battery assembly and the housing assembly, thereby effectively improving the sound quality of the earphone.

[0023] It is understood that the foregoing general description and the following detailed description are exemplary and explanatory only and are not intended to limit the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] In order to more clearly illustrate the technical solutions of the embodiments of the present disclosure, the accompanying drawings required to be used in the description of the embodiments are briefly described below. Obviously, the accompanying drawings in the following description are only some examples or embodiments of the present disclosure, and it is possible for those having ordinary skills in the art to obtain other drawings in accordance with these drawings without creative labor.

[0025] FIG. 1 is a schematic diagram illustrating a front contour of an ear of a user;

[0026] FIG. 2 is a schematic diagram illustrating an overall structure of an earphone according to some embodiments of the present disclosure;

[0027] FIG. 3 is a schematic diagram illustrating a structure of a section A of the earphone in FIG. 2;

[0028] FIG. 4 is a schematic diagram illustrating a structure of a shell in FIG. 3;
[0029] FIG. 5 is a partial cross-sectional view of the earphone in FIG. 3 along a B-direction; and
[0030] FIG. 6 is a partial cross-sectional view of a portion C of the earphone in FIG. 3.

DETAILED DESCRIPTION

[0031] The present disclosure is further described in detail below with reference to the accompanying drawings and embodiments. It is particularly noted that the following embodiments are only used to illustrate the present disclosure, but are not intended to limit the scope of the present disclosure. Similarly, the following embodiments are only some embodiments of the present disclosure rather than all embodiments, and all other embodiments obtained by those having ordinary skills in the art without creative labor are within the scope of protection of the present disclosure.

[0032] Reference to “embodiment” in the present disclosure means that a specific feature, structure, or characteristic described with reference to the embodiment may be included in at least one embodiment of the present disclosure. It is explicitly and implicitly understood by those skilled in the art that the embodiments described in the present disclosure may be combined with other embodiments.

[0033] Referring to FIG. 1, an ear **100** of a user includes an external ear canal (i.e., ear canal) **501**, a cavity of auricular concha **502**, a cymba conchae **503**, a triangular fossa **504**, an antihelix **505**, a scapha **506**, a helix **507**, a tragus **508**, and other physiological parts. Although the external ear canal **501** has a certain depth and extends to an eardrum of the ear, for the ease of description and based on FIG. 1, the external ear canal **501** specifically refers to an entrance (i.e., an earhole) of the external ear canal **501** away from the eardrum unless otherwise specified in the present disclosure. Furthermore, the cavity of auricular concha **502**, the cymba conchae **503**, the triangular fossa **504**, and other physiological parts have a certain volume and depth; and the cavity of auricular concha **502** is directly connected with the external ear canal **501**, i.e., it can be simply regarded as that the earhole is located at a bottom of the cavity of auricular concha **502**.

[0034] Furthermore, different users may have individual differences, resulting in different shapes, sizes, and other dimensional differences in the ears. For ease of description and to reduce (or even eliminate) the individual differences among different users, a simulator containing the head and (left and right) ears is prepared based on ANSI: S3.36, S3.25 and IEC: 60318-7 standards, such as GRAS 45BC KEMAR. Accordingly, in the present disclosure, descriptions such as “the user wears the earphone **100**”, “the earphone **100** is in a wearing state” and “in a wearing state” refer to the earphone **100** described in the present disclosure being worn on the ear of the simulator. Precisely because different users have individual differences, there may be certain differences between the earphone **100** worn by different users and the earphone **100** worn on the ear of the simulator, but such differences should be tolerated.

[0035] Referring to FIGS. 1-4, the earphone **100** may include a core module **1** and an ear hook **2**. The core module **1** may include a housing assembly **10**, a battery assembly **20**, and a driving assembly **30**. The ear hook **2** may be hook-shaped, and the earphone **100** may be worn, through the ear hook **2**, at a position near an ear canal of an ear of a user but not blocking the ear canal. The ear hook **2** may be connected with the housing assembly **10** by snap-fitting with glue, or by threading. In a wearing state, the core module **1** may be located on a front side of the ear, and the ear hook **2** may be at least partially located on a rear side of the ear of the user, so that the earphone **100** is hung on the ear. In some embodiments, the housing assembly **10** is also referred to as a housing.

[0036] Referring to FIG. 3, FIG. 4, and FIG. 5, the housing assembly **10** may form an accommodation space, and the accommodation space may include a first accommodation space **121a** and a second accommodation space **122a**. The battery assembly **20** may be disposed along a first preset assembly direction **Z1** and accommodated in the first accommodation space **121a**, and the driving assembly **30** may be disposed along a second preset assembly direction **Z2** and accommodated in the second accommodation space **122a**. The driving assembly **30** may include a

loudspeaker for providing an air-conducted sound to the external ear canal of the ear of the user. The battery assembly **20** may include a top wall surface and a bottom wall surface spaced apart in a thickness direction (not shown in the figure) and a peripheral wall surface **21** connecting the top wall and the bottom wall.

[0037] Specifically, in one embodiment, the housing assembly **10** includes a first shell **11** and a second shell **12**, and the first shell **11** and the second shell **12** are connected with each other and buckled with each other to form the accommodation space (including the first accommodation space **121a** and the second accommodation space **122a**). The battery assembly **20** and the driving assembly **30** are respectively accommodated in the accommodation space. That is, the battery assembly **20** is accommodated in the first accommodation space **121a**, and the driving assembly **30** is accommodated in the second accommodation space **122a**.

[0038] The first preset assembly direction **Z1** and the second preset assembly direction **Z2** may be provided at an acute angle. In other words, an intersection angle θ of a projection of a positive direction of the second preset assembly direction **Z2** in a reference plane and a projection of a positive direction of the first preset assembly direction **Z1** in the reference plane may be an acute angle. In some embodiments, the reference plane is a plane parallel to or coplanar with a plane where the first preset assembly direction **Z1** and the second preset assembly direction **Z2** are located. In some embodiments, the reference plane is an XY-plane formed by an X-direction and a Y-direction as shown in FIG. 3, i.e., a plane formed by a long axis direction and a thickness direction of the core module **1**. Optionally, the intersection angle of the projection of the positive direction of the second preset assembly direction **Z2** in the reference plane and the projection of the positive direction of the first preset assembly direction **Z1** in the reference plane is located on a side of the housing facing the ear of the user. In one embodiment, the intersection angle θ between the projection of the positive direction of the second preset assembly direction **Z2** in the reference plane and the projection of the positive direction of the first preset assembly direction **Z1** in the reference plane is in a range of 10° - 14° or in a range of 11° - 14° . For example, the intersection angle θ is 10° , 10.5° , 11° , 11.5° , 12° , 12.5° , 13° , 13.5° , or 14° . In this case, a tilt degree of the driving component **30** relative to the battery assembly **20** is large, and a length of a projection of the driving component **30** in a first direction **X** is reduced, which facilitates reducing a length of the housing component **10** in the first direction **X**, reducing a volume of the core module, and making the earphone tend to miniaturize as a whole smaller, which is easier for the user to carry and wear, and improves user experience.

[0039] In some embodiments, referring to FIGS. 3-5, in some embodiments, a first limiting portion **1211** is provided in the first accommodation space **121a**. A projection of the first limiting portion **1211** on a cross section perpendicular to the first preset assembly direction **Z1** may not overlap with a projection of the battery assembly **20** on the cross section perpendicular to the first preset assembly direction **Z1**, and the first limiting portion **1211** may be configured to limit a position of the battery assembly **20**. Specifically, the first limiting portion **1211** may be provided on a side of the battery assembly **20** along a direction perpendicular to the first preset assembly direction **Z1**, so that the first limiting portion **1211** may abut against the peripheral wall surface **21** of the battery assembly **20** parallel to the first preset assembly direction **Z1**. Accordingly, the first limiting portion **1211** can effectively limit the movement of the battery assembly **20** relative to the housing assembly **10** in a plane perpendicular to the first preset assembly direction **Z1**, which effectively improves the stability of the connection between the battery assembly **20** and the housing assembly **10**, and effectively prevents the noise generated by the relative movement of the battery assembly **20** and the housing assembly **10** during operation of the earphone **100**, thereby effectively improving the sound quality of the earphone **100** and improving the listening effect of the user.

[0040] In some embodiments, referring to FIG. 3, FIG. 4, and FIG. 5, the first limiting portion **1211** is configured at least to limit rotation of the battery assembly **20** around the first preset assembly direction **Z1** in a plane perpendicular to the first preset assembly direction. For example, the

peripheral wall surface **21** of the battery assembly **20** is parallel to the first preset assembly direction **Z1** (or in some embodiments, an angle θ is formed between the peripheral wall surface **21** and the first preset assembly direction **Z1**, and the angle θ is in a range of) 0° - 5° , and a portion of the peripheral wall surface **21** protrudes in the direction perpendicular to the first preset assembly direction **Z1** to form a limiting protrusion. Merely by way of example, when the battery assembly **20** is disposed in the first accommodation space **121a** along the first preset assembly direction **Z1**, the limiting protrusion on the peripheral wall surface **21** abuts against the first limiting portion **1211** to effectively limit the rotation of the battery assembly **20** around the first preset assembly direction **Z1**, which effectively improves the stability of the connection between the battery assembly **20** and the housing assembly **10**, and effectively prevents the noise generated by the relative movement of the battery assembly **20** and the housing assembly **10** during operation of the earphone **100**, thereby effectively improving the sound quality of the earphone **100** and improving the listening effect of the user. In some embodiments, based on the above embodiment, while at least one side of the first limiting portion **1211** parallel to the first preset assembly direction **Z1** abuts against a side of the limiting protrusion parallel to the first preset assembly direction **Z1**, the first limiting portion **1211** extends in the direction perpendicular to the first preset assembly direction **Z1** to the peripheral wall surface **21** of the battery assembly **20** and abuts against the peripheral wall surface **21**.

Accordingly, the first limiting portion **1211** is not only configured to limit the rotation of the battery assembly **20** around the first preset assembly direction **Z1** in the plane perpendicular to the first preset assembly direction **Z1**, but also plays a certain limiting role in the movement of the battery assembly **20** relative to the housing assembly **10** in the plane perpendicular to the first preset assembly direction **Z1**, thereby further improving the stability of the connection between the battery assembly **20** and the housing assembly **10**.

[0041] In some embodiments, referring to FIG. 3, FIG. 4, and FIG. 5, the battery assembly **20** has a thickness direction (not shown in the figure), and an angle between the first preset assembly direction **Z1** and the thickness direction of the battery assembly **20** may be in a range of 0° - 5° . Specifically, the battery assembly **20** may have the thickness direction, and the angle between the first preset assembly direction **Z1** and the thickness direction of the battery assembly **20** may be in a range of 0° - 5° , so that a size of the housing assembly **10** in a second direction **Y** can be effectively reduced. In one embodiment, the first preset assembly direction **Z1** and the thickness direction of the battery assembly **20** overlap, i.e., the angle between the first preset assembly direction **Z1** and the thickness direction of the battery assembly **20** is set to 0° . Accordingly, the size of the housing assembly **10** in the second direction **Y** can be made smaller, thereby further reducing the size of the core module and improving the comfort of the user.

[0042] Referring to FIG. 3, FIG. 4, and FIG. 5, the bottom wall surface of the battery assembly **20** may abut against a first bottom wall of the first accommodation space **121a**. The first bottom wall of the first accommodation space **121a** refers to an inner wall surface of the second shell facing the user in the wearing state. A certain amount of adhesive may be laid on the first bottom wall for bonding the battery assembly **20**. In some embodiments, the first limiting portion **1211** abuts against the peripheral wall surface **21** of the battery assembly **20** under the premise that the battery assembly **20** abuts against the first bottom wall. In this way, when the battery assembly **20** and the housing assembly **10** are fixedly connected by the adhesive such as glue, the first limiting portion **1211** may provide more bonding area between the housing assembly **10** and the battery assembly **20**, thereby effectively improving the connection strength between the battery assembly **20** and the housing assembly **10**, and preventing the battery assembly **20** from vibrating in the housing assembly **10**.

[0043] In some embodiments, referring to FIG. 3, FIG. 4, and FIG. 5, the first accommodation space **121a** includes a second limiting portion **1212**, and the second limiting portion **1212** is configured to limit movement of the battery assembly **20** along a radial direction of the battery assembly **20**. That is, the second limiting portion **1212** may be configured to limit displacement

movement of the battery assembly **20** relative to the housing assembly **10** in the direction perpendicular to the first preset assembly direction **Z1**. It can also be understood that the second limiting portion **1212** can increase abutment points on the battery assembly **20** in the plane perpendicular to the first preset assembly direction **Z1**, thereby reducing the possibility of the movement of the battery assembly **20** relative to the housing assembly **10** in the plane perpendicular to the first preset assembly direction **Z1**, and effectively improving the stability of the connection between the battery assembly **20** and the housing assembly **10**. In some embodiments, the second limiting portion **1212** is disposed at a position opposite to the first limiting portion **1211**. The first limiting portion **1211** limits the rotation of the battery assembly **20** in a manner described in the above embodiments, and the second limiting portion **1212** abuts against the peripheral wall surface **21** of the battery assembly **20** at the position opposite to the first limiting portion **1211**, thereby effectively limiting the movement of the battery assembly **20** relative to the housing assembly **10** in the direction perpendicular to the first preset assembly direction **Z1**. [0044] Referring to FIG. 3, FIG. 4, and FIG. 5, the first accommodation space **121a** may include a first inner sidewall (not shown in the figure), and the first inner sidewall may be parallel to an inner wall of the housing in the thickness direction of the core module. In some embodiments, the first inner sidewall is parallel to the first preset assembly direction **Z1**, where the second limiting portion **1212** and the first inner sidewall are an integral structure, or the second limiting portion **1212** is a limiting protrusion disposed on the first inner sidewall, or a combination of the integral structure and the limiting protrusion. Taking the second limiting portion **1212** and the first inner sidewall as an integral structure as an example, a portion of the first inner sidewall protrudes in the direction perpendicular to the first preset assembly direction **Z1** to form the second limiting portion **1212**, and the second limiting portion **1212** abuts against the peripheral wall surface **21** of the battery assembly **20**, thereby limiting the movement of the battery assembly **20** relative to the housing assembly **10** in the direction perpendicular to the first preset assembly direction **Z1**. In some embodiments, a limiting protrusion is provided on the first inner sidewall as the second limiting portion **1212**, which is not elaborated here.

[0045] In some embodiments, a positioning portion **1213** is disposed in the first accommodation space **121a**; the positioning portion **1213** and the first limiting portion **1211** are spaced apart, the battery assembly **20** is provided with a positioning protrusion **22**, and when being used for the battery assembly **20** is assembled along the first preset assembly direction **Z1**, the positioning protrusion **22** is disposed between the positioning portion **1213** and the first limiting portion **1211**, so that the positioning portion **1213** and the first limiting portion **1211** perform assembly positioning on the battery assembly **20** when the battery assembly **20** is assembled in the first accommodation space **121a** along the first preset assembly direction **Z1**.

[0046] Specifically, referring to FIG. 4 and FIG. 5, the first accommodation space **121a** may be provided with the positioning portion **1213**, and the battery assembly **20** may be provided with the positioning protrusion **22**, and the positioning protrusion **22** may be disposed on the peripheral wall surface **21** of the battery assembly **20**. Merely by way of example, when the battery assembly **20** is assembled in the first accommodation space **121a** of the housing assembly **10** along the first preset assembly direction **Z1**, a guide space along the first preset assembly direction **Z1** is formed between the first limiting portion **1211** and the positioning portion **1213**, and the positioning protrusion **22** is placed in the guide space. In other words, the battery assembly **20** can be quickly assembled in the first accommodation space **121a** along the first preset assembly direction **Z1** after performing positioning by the positioning protrusion **22** and the guide space, thereby effectively improving the assembly efficiency of the battery assembly **20**. In some embodiments, the positioning protrusion **22** is used as the limiting protrusion to position the battery assembly **20** during assembly while limiting the movement of the battery assembly **20** in the housing assembly **10**. In this way, the complexity of the battery assembly **20** can be reduced and the production cost can be reduced.

[0047] In some embodiments, the core module **1** includes an electrode member **40** electrically connected with the battery assembly **20**. The electrode member **40** may be disposed in the positioning portion **1213** and extend from the first accommodation space **121a** to outside of the housing assembly **10**. Specifically, in one embodiment, the electrode member **40** is a component electrically connected with the battery assembly **20**, and the positioning portion **1213** is provided with an electrode mounting portion connecting the first accommodation space **121a** and an external space of the housing assembly **10**, wherein the electrode member **40** is disposed in the electrode mounting portion and extends along the electrode mounting portion to the outside of the housing assembly **10**, and the battery assembly **20** can be electrically connected through the electrode member **40** with an external charging case of the earphone **100** or other charging devices to realize a charging function. In one embodiment, a mounting hole for mounting the electrode member is provided on the positioning portion **1213**, and a corresponding through hole is provided on the housing component **10**, so that after the electrode member **40** is disposed in the mounting hole on the positioning portion **1213**, the electrode member can directly extend out of the housing assembly **10** through the through hole to be electrically connected with a charging device such as the charging case. In this way, the processing difficulty of the housing assembly **10** can be reduced and the cost can be reduced.

[0048] Two positioning portions **1213** may be provided. Referring to FIG. 5, the two positioning portions **1213** may be respectively disposed on two sides of the first limiting portion **1211**, and the positioning protrusion **22** may be disposed between the first limiting portion **1211** and any one of the two positioning portions **1213**. In some embodiments, two corresponding electrode members **40** may be provided, and the two electrode members **40** may be disposed on the two corresponding positioning portions **1213**, respectively.

[0049] In some embodiments, the battery assembly **20** leads a wire for electrically connecting with other electronic components of the core module **1**. The first accommodation space **121a** may include a first bottom wall (not shown in the figure). The first bottom wall refers to an inner wall surface of the second shell facing the user in the wearing state. A lead wire groove **1214** may be disposed on the first bottom wall to accommodate the wire. Specifically, the first bottom wall may be provided with an abutting portion that abuts against the bottom wall surface of the battery assembly **20**, and the lead wire groove **1214** to accommodate the wire. The lead wire groove **1214** is disposed on the first bottom wall to reasonably arrange the wire, thereby effectively reducing the space occupied by the wire, and effectively reducing the spatial volume of the housing assembly **10**.

[0050] In some embodiments, the housing assembly **10** is provided with a partition wall **1215**. The partition wall **1215** may be configured to separate the first accommodation space **121a** and the second accommodation space **122a**, and the first limiting portion **1211** may be convexly disposed on a side of the partition wall **1215** facing the first accommodation space **121a**. Specifically, the partition wall **1215** may be configured to separate the first accommodation space **121a** and the second accommodation space **122a** from each other, so that the battery assembly **20** and the driving assembly **30** can be effectively separated, thereby preventing irreparable damage to the driving assembly **30** caused by leakage of battery fluid from the battery assembly **20**, or the like.

[0051] In some embodiments, the battery assembly **20** and the driving assembly **30** are disposed in the second shell **12**, the second shell **12** is provided with an open end, and a plurality of positioning grooves **1231** are provided at an edge close to the open end of the second shell **12**. The first shell **11** may be configured to cover the second shell **12**. An edge close to the first shell **11** may be provided with a plurality of positioning columns **112** corresponding to the plurality of positioning grooves **1231**. At least a portion of the plurality of positioning columns **112** may be embedded in the plurality of positioning grooves **1231**. Notches of the plurality of positioning grooves **1231** may be provided with first inclined wall surfaces **1232**. The first inclined wall surfaces **1232** may be configured to guide the plurality of positioning columns **112** to be embedded into the plurality of

positioning grooves **1231**.

[0052] Specifically, referring to FIGS. **3-6**, the first shell **11** may be provided with a third accommodation cavity **111**, and the second shell **12** may be provided with a first sub-accommodation cavity **121b** and a second sub-accommodation cavity **122b** separated by the partition wall **1215**. After the first shell **11** and the second shell **12** are buckled and connected, the third accommodation cavity **111** may be communicated with the first sub-accommodation cavity **121b** and the second sub-accommodation cavity **122b**, respectively, to form the first accommodation space **121a** and the second accommodation space **122a**, respectively, wherein the first accommodation space **121a** and the second accommodation space **122a** are collectively referred to as the accommodation space of the housing assembly **10**. In this way, the second shell **12** may serve as an accommodation body for accommodating the battery assembly **20** and the driving assembly **30**. The battery assembly **20** may be accommodated in the first sub-accommodation cavity **121b**, and the driving assembly **30** may be accommodated in the second sub-accommodation cavity **122b**. The first shell **11** may serve as a cover body to cover the second shell **12**, so that the battery assembly **20** and the driving assembly **30** are respectively enclosed in the first accommodation space **121a** and the second accommodation space **122a**.

[0053] In some embodiments, the first shell **11** and the second shell **12** are positioned and assembled by the plurality of positioning columns **112** and the plurality of positioning grooves **1231**, thereby effectively improving the assembly efficiency of the first shell **11** and the second shell **12**. In addition, since the second shell **12** serves as a main body for accommodating the battery assembly **20** and the driving assembly **30**, it is not convenient to move after the battery assembly **20** and the driving assembly **30** are assembled. Accordingly, when the first shell **11** and the second shell **12** are assembled, the second shell **12** remains as still as possible (e.g., the second shell **12** provided with the battery assembly **20** and the driving assembly **30** is fixed by a positioning base), and then the first shell **11** covers the second shell **12** from top to bottom along the direction of gravity to complete the assembly. The plurality of positioning grooves **1231** may be disposed on the second shell **12**, and the plurality of positioning columns **112** may be disposed on the first shell **11**, so that when the first shell **11** and the second shell **12** are assembled, the plurality of positioning grooves **1231** are located below the plurality of positioning columns **112**, and an assembler can more easily complete the alignment of the plurality of positioning columns **112** and the plurality of positioning grooves **1231**. In addition, the notches of the plurality of positioning grooves **1231** may be provided with the first inclined wall surfaces **1232**, and the first inclined wall surfaces **1232** may be configured to guide the plurality of positioning columns **112** to be embedded into the plurality of positioning grooves **112**, which can effectively improve the smoothness of the plurality of positioning columns **112** being embedded into the plurality of positioning groove **1231**, thereby effectively improving the assembly efficiency of the first shell **11** and the second shell **12**. In this way, the second shell **12** provided with the battery assembly **20** and the driving assembly **30** does not need to be significantly adjusted, which effectively ensures the overall assembly accuracy of the earphone **100** and the stability of the assembly of the battery assembly **20** and the driving assembly **30**.

[0054] In some embodiments, the plurality of positioning columns **112** are configured to be formed along a first demolding direction **Z3** of the first shell **11**. The core module **1** may comprise a circuit board **50** accommodated in the accommodation space, and the first demolding direction **Z3** may be perpendicular to a plane where the circuit board **50** is located. In other words, an axis of the plurality of positioning columns **112** may be parallel to the first demolding direction **Z3**, i.e., the axis of the plurality of positioning column **112** may be perpendicular to the plane where the circuit board **50** is located. Specifically, referring to FIG. **3** and FIG. **6**, the first demolding direction **Z3** is a direction of demolding of the first shell **11** during processing. Setting a forming direction of the plurality of positioning columns **112** to be the same as the demolding direction of the first shell **11** effectively reduces the molding process of the first shell **11**, and reduces the process difficulty,

thereby effectively simplifying the structure of the first shell **11** and effectively improving the molding efficiency of the first shell **11**. Further, the circuit board **50** may be a control circuit element of the earphone **100** and configured to be provided between the first shell **11** and the second shell **12**. That is, the circuit board **50** may be located in the accommodation space to cross the first accommodation space **121a** and the second accommodation space **122a**. The circuit board **50** may be fixedly connected with the first shell **11**, and the plane where the circuit board **50** is located may be perpendicular to the first demolding direction **Z3**. In this way, the spatial volume occupied by the circuit board **50** in the second direction **Y** (wherein the first direction **X**, the first demolding direction **Z3**, and the second direction **Y** are located in the same plane, and the first direction **X** and the second direction **Y** are perpendicular to each other) is effectively reduced, thereby effectively reducing the overall size of the housing assembly **10** along the second direction **Y**, reducing the volume of the earphone **100**, and improving the wearing comfort of the user.

[0055] In some embodiments, the first shell **11** is provided with hot melt columns **113**, and the hot melt columns **113** are integrally formed with the first shell **11** along the first demolding direction **Z3**. The hot melt columns **113** may be configured to fix the circuit board **50**. Specifically, referring to FIG. 3, the circuit board **50** may be fixedly connected with the first shell **11** through the hot melt columns **113**. The hot melt columns **113** may be formed along the first demolding direction **Z3**, which can effectively reduce the molding process of the first shell **11**, thereby effectively improving the molding efficiency of the first shell **11**.

[0056] In some embodiments, the first preset assembly direction **Z1** and the second preset assembly direction **Z2** are set at an acute angle; an angle between the first demolding direction **Z3** and the first preset assembly direction **Z1** is in the range of 0° - 5° . Specifically, referring to the present disclosure above, the battery assembly **20** may be accommodated in the first accommodation space **121a** along the first preset assembly direction **Z1**, and the driving assembly **30** may be accommodated in the second accommodation space **122a** along the second preset assembly direction **Z2**. A relationship between the first preset assembly direction **Z1** and the second preset assembly direction **Z2** is as described above, which is not repeated here. In some embodiments, the first preset assembly direction **Z1** is parallel to the first demolding direction **Z3**, i.e., the angle between the first demolding direction **Z3** and the first preset assembly direction **Z1** is 0° . In this way, the overall demolding structure of the housing assembly **10** conforms to a combination relationship of the preset assembly directions (e.g., the first preset assembly direction **Z1** and the second preset assembly direction **Z2**) of the battery assembly **20** and the driving assembly **30** as described above, thereby effectively reducing the overall structural size of the housing assembly **10**. The angle range between the first demolding direction **Z3** and the first preset assembly direction **Z1** may be in a range of 0° - 5° , which can also achieve the above effect.

[0057] In some embodiments, as described above, the second shell **12** is provided with the first sub-accommodation cavity **121b** and the second sub-accommodation cavity **122b**, the battery assembly **20** is accommodated in the first sub-accommodation cavity **121b**, and the driving assembly **30** is accommodated in the second sub-accommodation cavity **122b**. Further, the circuit board **50** may cross the first sub-accommodation cavity **121b** and the second sub-accommodation cavity **122b**. In this way, a projection of the circuit board **50** along a direction of a normal line of a main surface of the circuit board **50** may be configured to cover at least part of the driving assembly **30** and the battery assembly **20** simultaneously, which can make full use of the space inside the earphone **100**, making the earphone **100** compact, thereby reducing the volume of the earphone **100** and making it easier for users to wear.

[0058] Further, the plurality of positioning grooves **1231** may be configured to be formed along a second demolding direction **Z4** of the second shell **12**. The first demolding direction **Z3** and the second demolding direction **Z4** may be set at an acute angle, and the acute angle may be in a range of 0° - 10° . Specifically, referring to FIGS. 3-6, the first preset assembly direction **Z1** may be parallel to the first demolding direction **Z3**, and the second preset assembly direction **Z2** may be

parallel to the second demolding direction **Z4**. In other words, the second preset assembly direction **Z2** may coincide with the second demolding direction **Z4**, which can reduce the molding process of the second shell **12**. The first demolding direction **Z3** and the second demolding direction **Z4** may be set at an acute angle, so that the overall demolding structure of the housing assembly **10** conforms to the combination relationship of the preset assembly directions (e.g., the first preset assembly direction **Z1** and the second preset assembly direction **Z2**) of the battery assembly **20** and the driving assembly **30** as described above, thereby effectively reducing the overall structural size of the housing assembly **10**.

[0059] In some embodiments, groove walls of the plurality of positioning grooves **1231** include the first inclined wall surfaces **1232** which are inclined compared to the second demolding direction **Z4** and are biased toward the first demolding direction **Z3**. The first inclined wall surfaces **1232** may be configured to guide the plurality of positioning columns **112** to be embedded into the plurality of positioning grooves **1231**.

[0060] Specifically, referring to FIG. 6, the positioning grooves **1231** may be provided with the first inclined wall surfaces **1232**. The first inclined wall surfaces **1232** may be inclined compared to the second demolding direction **Z4** and biased toward the first demolding direction **Z3**, and an inclination angle of the first inclined wall surfaces **1232** relative to the second demolding direction **Z4** may be the same as the angle between the first demolding direction **Z3** and the second demolding direction **Z4**. In this way, the first inclined wall surfaces **1232** can effectively guide the plurality of positioning columns **112** to be embedded into the plurality of positioning grooves **1231**.

[0061] In some embodiments, the second shell **12** is integrally formed, and the plurality of positioning grooves **1231** are formed at the edge of the open end of the second shell **12**, so that the assembly process of the second shell **12** can be reduced and the assembly efficiency can be improved. In some embodiments, the second shell **12** includes a shell body **12A** and a bracket **12B**. The bracket **12B** and the shell body **12A** may be separately formed and assembled. The shell body **12A** may be provided with the first sub-accommodation space **121b** and the second sub-accommodation space **122b**. The bracket **12B** may be assembled in the second sub-accommodation space **122b** for supporting and assembling the driving assembly **30**. A portion of the positioning grooves **1231** may be disposed on the shell body **12A**, and the other portion of the positioning grooves **1231** may be disposed on the bracket **12B**, so as to facilitate flexible adjustment when the first shell **11** and the second shell **12** are buckled, and improve the accuracy when the first shell **11** and the second shell **12** are buckled.

[0062] In some embodiments, the first shell **11** is provided with a first support platform **114**, and the plurality of positioning columns **112** are disposed on the first support platform **114** and extend outward from a bottom surface of the first support platform **114**. The second shell **12** may be provided with a second support platform **123**, and the plurality of positioning grooves **1231** may be disposed on the second support platform **123** and form the notches on a top surface of the second support platform **123**. For example, when the first shell **11** and the second shell **12** are buckled, and at least a portion of the plurality of positioning columns **112** are embedded into the plurality of positioning grooves **1231**, a first gap **L1** is formed between the bottom surface of the first support platform **114** and the top surface of the second support platform **123**, in this case, a second gap **L2** is formed between edges of the first shell **11** and the second shell **12** which are buckled with each other. A minimum gap height of the first gap **L1** may be less than a minimum gap height of the second gap **L2**. The second gap **L2** may be configured to accommodate a fixing adhesive. By setting the minimum gap height of the first gap **L1** to be less than the minimum gap height of the second gap **L2**, after the first shell **11** and the second shell **12** are assembled in place, the second gap **L2** still has sufficient height to accommodate the adhesive such as glue, thereby effectively ensuring the effective sealed connection between the edges of the first shell **11** and the second shell **12**.

[0063] In some embodiments, a minimum gap distance of the first gap **L1** is in a range of 0.02-0.07

mm, and a minimum gap distance of the second gap L2 is in a range of 0.08-0.12 mm.
[0064] The above descriptions are only some embodiments of the present disclosure, and do not limit the protection scope of the present disclosure. Any equivalent device or equivalent process transformation made using the contents of the specification and the drawings of the present disclosure, or directly or indirectly used in other related technical fields, are also included in the patent protection scope of the present disclosure.

Claims

1. A core module, comprising: a housing provided with a first accommodation space and a second accommodation space, a battery assembly accommodated in the first accommodation space along a first preset assembly direction; and a driving assembly accommodated in the second accommodation space along a second preset assembly direction, wherein the first preset assembly direction and the second preset assembly direction are provided at an acute angle, a first limiting portion is disposed in the first accommodation space, a projection of the first limiting portion on a cross section perpendicular to the first preset assembly direction does not overlap with a projection of the battery assembly on the cross section perpendicular to the first preset assembly direction, the first limiting portion is configured to limit a position of the battery assembly, and the first limiting portion is configured to stop the battery assembly at least in a direction perpendicular to the first preset assembly direction to limit rotation of the battery assembly around the first preset assembly direction.
2. The core module of claim 1, wherein the battery assembly has a thickness direction, and an angle between the first preset assembly direction and the thickness direction of the battery assembly is in a range of 0° - 5° .
3. The core module of claim 2, wherein the battery assembly includes a top wall surface and a bottom wall surface which are spaced apart along the thickness direction and a peripheral wall surface connecting the top wall surface and the bottom wall surface, and the first limiting portion is configured to abut against the peripheral wall surface of the battery assembly for stopping.
4. The core module of claim 1, wherein the first accommodation space includes a second limiting portion, and the second limiting portion is configured to limit movement of the battery assembly along a radial direction of the battery assembly.
5. The core module of claim 4, wherein the first accommodation space includes a first inner sidewall, wherein the second limiting portion is a portion of the first inner sidewall.
6. The core module of claim 5, wherein the second limiting portion is a limiting protrusion disposed on the first inner sidewall.
7. The core module of claim 1, wherein a positioning portion is disposed in the first accommodation space, the positioning portion and the first limiting portion are spaced apart, the battery assembly is provided with a positioning protrusion, and the positioning protrusion is disposed between the positioning portion and the first limiting portion along the first preset assembly direction, so that the positioning portion and the first limiting portion perform assembly positioning on the battery assembly when the battery assembly is assembled in the first accommodation space along the first preset assembly direction.
8. The core module of claim 7, wherein the core module includes an electrode member electrically connected with the battery assembly, the electrode member is disposed in the positioning portion and extends from the first accommodation space to outside of the housing.
9. The core module of claim 8, wherein two positioning portions are provided, and the two positioning portions are respectively disposed on two sides of the first limiting portion, and the positioning protrusion is disposed between the first limiting portion and any one of the two positioning portions.
10. The core module of claim 8, wherein the battery assembly leads a wire, the first

accommodation space includes a first bottom wall, and a lead wire groove is disposed on the first bottom wall to accommodate the wire.

11. The core module of claim 1, wherein the housing is provided with a partition wall, the partition wall is configured to separate the first accommodation space and the second accommodation space, and the first limiting portion is convexly disposed on a side of the partition wall facing the first accommodation space.

12. The core module of claim 1, wherein the housing includes a first shell and a second shell, the second shell is provided with an open end, the first shell and the second shell are connected with each other and the first shell covers the open end to form the first accommodation space and the second accommodation space, the second shell is provided with a plurality of positioning grooves, the first shell is provided with a plurality of positioning columns corresponding to the plurality of positioning grooves, at least a portion of the plurality of positioning columns are embedded in the plurality of positioning grooves, notches of the plurality of positioning grooves are provided with first inclined wall surfaces, and the first inclined wall surfaces are configured to guide the plurality of positioning columns to be embedded into the plurality of positioning grooves.

13. The core module of claim 12, wherein the plurality of positioning columns are configured to be formed along a first demolding direction of the first shell, the core module comprises a circuit board, and the first demolding direction is perpendicular to a plane where the circuit board is located.

14. The core module of claim 13, wherein an angle between the first demolding direction and the first preset assembly direction is in a range of 0° - 5° .

15. The core module of claim 13, wherein the plurality of positioning grooves are configured to be formed along a second demolding direction of the second shell, the first demolding direction and the second demolding direction are set at an acute angle.

16. The core module of claim 15, wherein the first inclined wall surfaces are inclined compared to the second demolding direction and are biased toward the first demolding direction.

17. The core module of claim 12, wherein the first shell is provided with a first support platform, the plurality of positioning columns are disposed on the first support platform and extend outward from a bottom surface of the first support platform, the second shell is provided with a second support platform, the plurality of positioning grooves are disposed on the second support platform and form the notches on a top surface of the second support platform.

18. The core module of claim 17, wherein when the at least a portion of the plurality of positioning columns are embedded in the plurality of positioning grooves, a first gap is formed between the bottom surface of the first support platform and the top surface of the second support platform, a second gap is provided between edges of the first shell and the second shell which are buckled with each other, a minimum gap height of the first gap is less than a minimum gap height of the second gap, and the second gap being configured to accommodate a fixing adhesive.

19. The core module of claim 18, wherein a minimum gap distance of the first gap is in a range of 0.02-0.07 mm, and a minimum gap distance of the second gap is in a range of 0.08-0.12 mm.

20. An earphone, comprising: the core module of claim 1; and an ear hook configured to be connected with the housing to make the core module be worn near an ear canal of a user but not block the ear canal.
